VLM-BASED AI-OCR · SHOP-FLOOR DATA COLLECTION

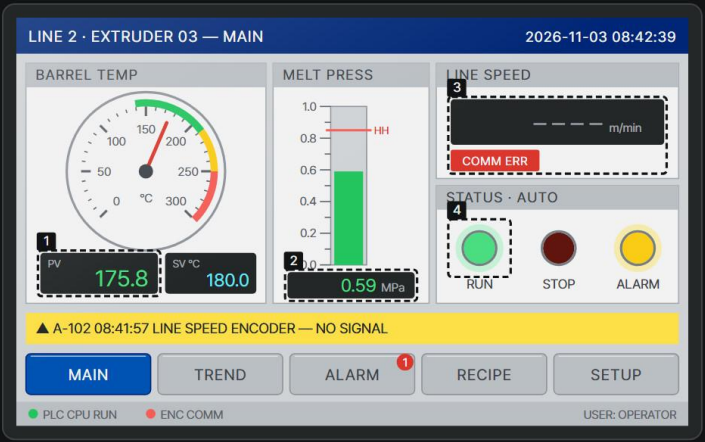
IntCloud OCR

The gauge numbers people used to read and note down, now read by a camera and kept as a record

Instrument monitors · status boards · PLC/HMI screens · gauges → item · value · unit · status

VLM: an AI model that understands images and text together · Equipment unchanged: a camera shoots the screen

SCREEN PLC / HMI screen



VLM →

DATA Extracted data

Item	Value	Unit	Status
1 Temperature	175.8	°C	Normal
2 Pressure	0.59	MPa	Normal
3 Line speed	—	m/min	Missing
4 Run status	RUN	—	Running

Read at 08:42:39 · reading interval 10 s

HISTORY Reading log · every reading is kept with its time



IntCloud OCR concept: a screen read into item · value · unit · status and kept with its time · illustrative screen and values

02 — OVERVIEW

Turn the screens of equipment that can’t connect into data

IntCloud OCR photographs the displays people used to read by eye and reads their values. Each value gets its item, unit and status, so you can look up past values and see the trend right away.

FIG. 01 READING EXAMPLE



IntCloud OCR reading example · PLC / HMI
An illustrative screen

01 WHAT IT READS

Displays people used to read by eye

A camera captures the values on instrument monitors, production status boards, PLC/HMI screens and gauges.

02 WHAT STAYS THE SAME

Equipment and existing systems

Because it works from camera images of the screen, the equipment and its existing systems stay unchanged.

03 WHAT YOU KEEP

Data with item, value, unit and status

Each reading is stored with its item, value, unit and status, ready for history tracking and trend analysis.

5 types
Displays it can read
Monitors · boards · HMI · gauges

4 steps
How a reading works
From capture to validation

4 fields
Attached to each value
Item · value · unit · status

10 s
Example reading interval
Or 30 s, 1 min, as needed

The numbers people wrote down, read by a camera and kept.

The equipment stays as it is; a camera turns the values on its screen into data.

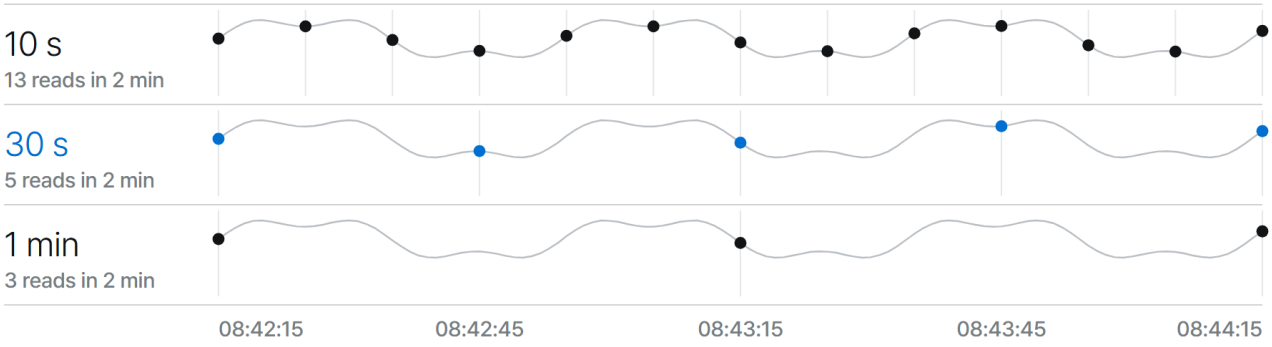
03 — HOW IT READS · READING LOG

One row for every reading, and the rows become a history

IntCloud OCR reads the screen at the interval you set and stores each value with the time it was read. The reading interval can be set to suit the site, such as 10 s, 30 s or 1 min.

FIG. 02 READING INTERVAL · HISTORY

Temperature · °C Grey line: the value on the screen · dots: the values IntCloud OCR read



Reading log Every 30 s · one row is added with each reading

Read at	Temperature · °C	Pressure · MPa	Line speed · m/min	Run status
08:42:15	181.0 Normal	0.62 Normal	— Missing	RUN Running
08:42:45	177.5 Normal	0.61 Normal	— Missing	RUN Running
08:43:15	179.9 Normal	0.62 Normal	— Missing	RUN Running
08:43:45	184.6 Normal	0.64 Normal	— Missing	RUN Running

Values read at each interval, and the log built every 30 s · values from the reading example (illustrative)

What is kept with each value

From the reading example · an illustrative screen

VALUE · UNIT

Name and unit attached

Not just ‘175.8’, but ‘Temperature 175.8 °C’: what was measured is stored with the value.

MISSING

Blank means ‘Missing’

If a communication error leaves the line speed blank, it is logged as ‘Missing’, not a number.

STATUS

Lamps and messages too

The RUN lamp is read as ‘Running’, and the ‘Alert!’ message on the instrument monitor as ‘Warning’.

Find past values. See the change.

Each item builds up with its time, ready for history tracking and trend analysis.

04 — COMPARE · CONVENTIONAL OCR

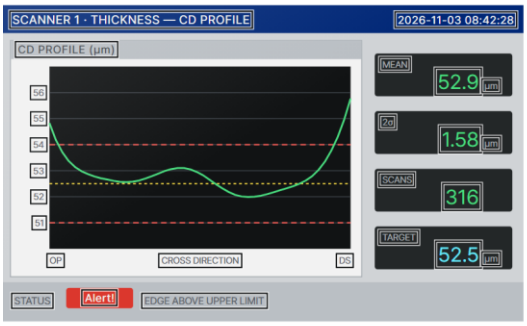
Reading characters, or understanding the screen

Conventional OCR reads one character at a time, and people must write rules to say which number means what. IntCloud OCR understands the whole screen and pulls out item, value, unit and status together.

FIG. 03 CONVENTIONAL OCR · VLM

Conventional OCR

Character by character · ROIs set for each screen



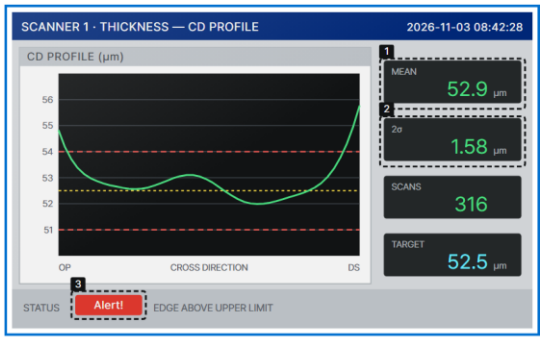
Output · characters in a row, nothing more

SCANNER 1 · THICKNESS — CD PROFILE
2026-11-03 08:42:28 / CD PROFILE (µm)
51 / 52 / 53 / 54 / 55 / 56 / OP

Which value is which needs separate rules

IntCloud OCR

Understands the whole screen in context



Output · item · value · unit · status

1 Measured value	52.9	µm	Normal
2 Statistic (2σ)	1.58	µm	Normal
3 Status message	Alert!	—	Warning

Left: character by character · right: the whole screen · the reading example's instrument monitor (illustrative screen)

How it differs from conventional OCR Product page 01 COMPARE

Conventional OCR

- Recognizes text character by character, so read positions (ROIs) and post-processing rules are set for each screen.
- Needs reconfiguring when the layout, font, or display format changes.
- Linking a value to its meaning (item, unit, status) takes separate rules.

IntCloud OCR

Built on a recent VLM (vision-language model)

- Understands the whole screen as an image in context and extracts item names, values, units, and status together.
- Interprets tables, graph panels, and status messages such as 'Alert', 'NO DATA', and 'Line Stopped'.
- You define the fields to extract (a schema), which makes adding or changing screen types easier.

From reading characters to reading meaning.

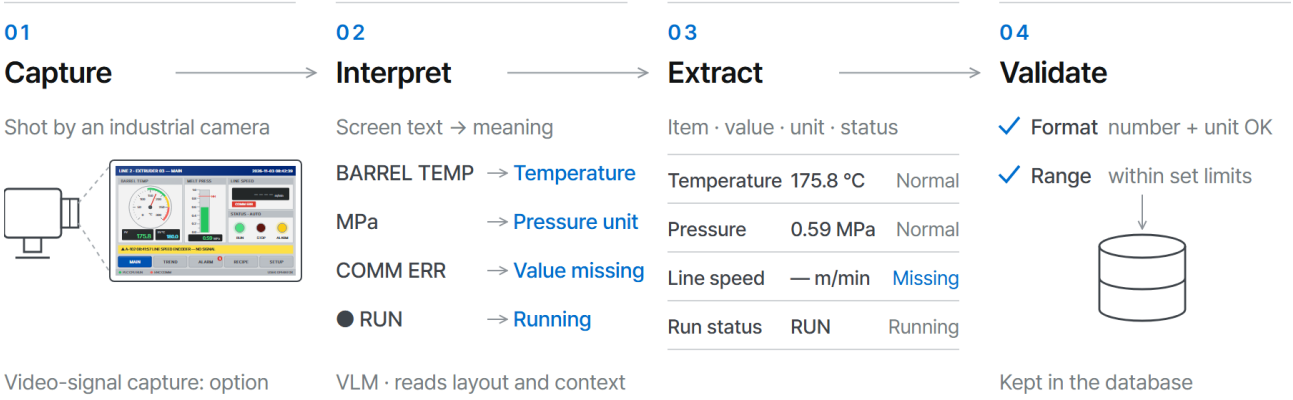
Define the fields to read (a schema), and screen types are easy to add or change.

05 — FLOW · CAPTURE TO DATABASE

Capture, understand, extract, check, then store

It starts with a camera photographing the screen. The VLM reads what the screen means, pulls the values out as a table, checks their format and range, then stores them in the database.

FIG. 04 CAPTURE · READ · EXTRACT · CHECK



How a reading works · concept diagram drawn on the reading example's touch panel (illustrative screen)

How a reading works

Product page 02 FLOW

01	Capture	An industrial camera captures the screen or gauge. Taking the video signal directly through a capture device is another option to review.
02	Interpret	The VLM understands the layout and context of the screen.
03	Extract	Item names, values, units and status come out as one row of a table.
04	Validate	Checks that the number and unit are in the right form (format) and inside the set range (range), then stores it in the database.

Add a camera. Leave the equipment as it is.

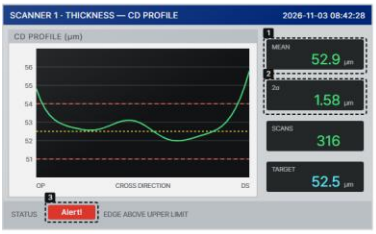
Because it works from camera images of the screen, the equipment and its existing systems stay unchanged.

06 — DISPLAYS · WHAT IT CAN READ

Status boards to dial gauges: five kinds of display it reads

The same camera and reading setup extends to more targets: the warning on an instrument monitor, the numbers on a status board, the lamps on an operator screen, 7-segment digits (drawn with seven bars) and where a gauge needle points.

FIG. 05 FIVE DISPLAYS



Instrument monitors

Measured values, statistics, and status messages on quality instruments such as QCS

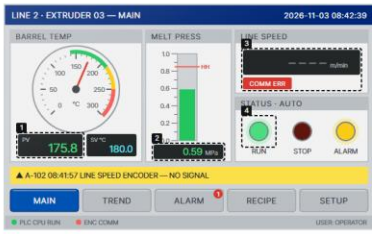
Measured value	52.9 µm	Normal
Statistic (2σ)	1.58 µm	Normal
Status message	Alert!	Warning



Production status boards

Output, targets, and run status on LED or monitor boards

Target	1,500 EA	
Output	1,252 EA	Normal
Run status	RUN	Running



PLC / HMI screens

Set points, current values, operating modes, and alarms on operator screens

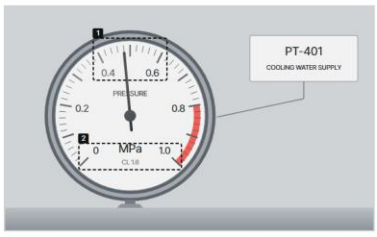
Temperature	175.8 °C	Normal
Pressure	0.59 MPa	Normal
Line speed	— m/min	Missing
Run status	RUN	Running



Digital gauges

Numbers and units on 7-segment and LCD displays

Present value	24.0 °C	Normal
Set value	24.0 °C	



Analog gauges

Scale and needle position on dial instruments such as pressure and temperature gauges

Reading	0.48 MPa	Normal
Scale range	0–1.0 MPa	

Getting started

01 Site survey
02 Define what to read
03 On-site validation
04 Rollout

The reading example's five displays and what was extracted from each · illustrative screens

* Reading accuracy depends on capture conditions and the display's specifications, so the scope is set for each target after on-site validation.

Accuracy is set after checking on site.

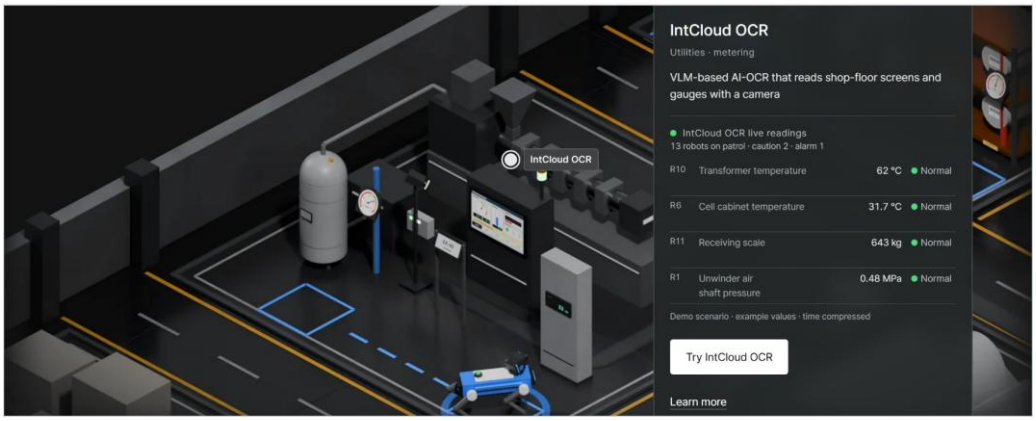
Site survey → define what to read → on-site validation → rollout: the scope is set for each target.

07 — LINEUP · RELATED PRODUCTS

Next to the inspection line, it gathers the plant's numbers

In the INTELLIZ product lineup, IntCloud OCR handles shop-floor data collection. In the factory digital twin of the Solution Experience, it reads the gauges and screens in the utilities and metering zone.

FIG. 06 SOLUTION EXPERIENCE · DA-01



Solution Experience · utilities zone · an example view of a virtual factory, not customer data

INTLEARN PRO · OCR

Text on parts: IntLearn Pro

Finding part markings and engraved text and reading them is the job of IntLearn Pro's OCR.

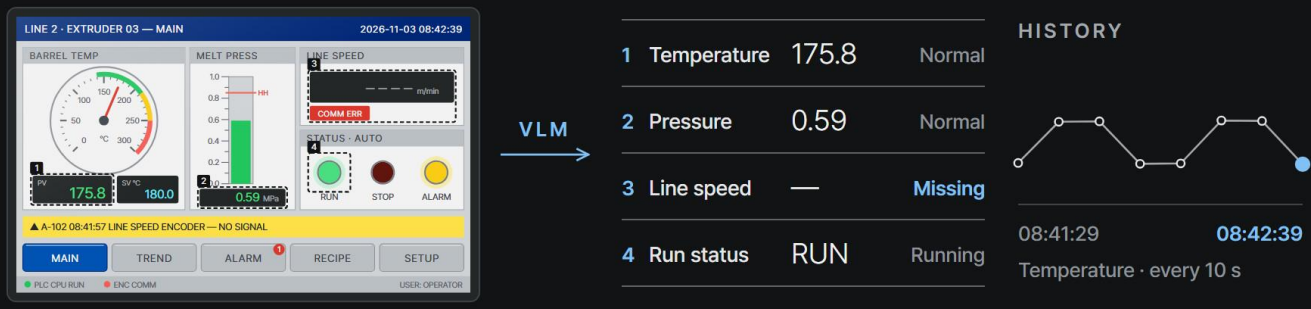
INTCLOUD OPTIC · DEMO SCENARIO

A robot patrols, IntCloud OCR files

IntCloud OCR files what a robot dog with IntCloud Optic reads as item, value, unit and status (example setup).

Product lineup intelliz.net/en/products/lineup

01	Physical AI inspection platform	IntCloud Optic	An AI optical inspection module that captures defects so they can be seen
		IntLearn Pro	Deep learning visual inspection software
		IntWise	Operate by speaking, and the know-how becomes a company asset
02	Physical AI platform	IntLearn Kinesis	Validate in a virtual factory before you install
03	Inspection runtime software	IntScan Pro	Runs line-scan camera inspection for continuous materials
		IntSight	Runs area-scan camera inspection for discrete parts
		IntViewer Pro	Monitors inspection results
04	Shop-floor data collection	**IntCloud OCR**	VLM-based AI-OCR that reads shop-floor screens and gauges with a camera
05	Process control	IntLearn ML	Machine learning that finds the cause of defects in process data and predicts setpoints



IntCloud OCR · from a shop-floor screen to a table and a record · illustrative example

08 — CONTACT · NEXT STEPS

Want to turn your screens into data?

Including a photo of the screen or gauge and the values you want to collect helps us prepare our answer.

01	A photo of the screen or gauge to read	02	Values to collect (e.g. temperature, pressure)
03	Display type (monitor, board, HMI, gauge)	04	Reading interval (e.g. 10 s, 30 s, 1 min)

Next steps

Where the product page leads · intelliz.net

Try it	INTELLIZ Solution Experience · IntCloud OCR in the factory digital twin (DA-01) intelliz.net/en/solutions/experience?station=da-01
Reading example	A reading example: pick one of five displays and read it intelliz.net/en/products/intcloud-ocr#try
Getting started	Site survey → define what to read → on-site validation → rollout intelliz.net/en/products/intcloud-ocr
Product lineup	INTELLIZ product lineup · shop-floor data collection intelliz.net/en/products/lineup
Email subject	[IntCloud OCR inquiry] sales@intelliz.net

CONTACT

sales@intelliz.net

+82-70-7006-8916

Seoul R&D Center #306C, 37, Seongsui-ro 22-gil, Seongdong-gu, Seoul, Republic of Korea
Main Office #305, 63, Buldang 36-gil, Seobuk-gu, Cheonan-si, Chungcheongnam-do, Republic of Korea
Manufacturing Lab B-403, Hyundai Silicon Alley Dongtan, 150, Dongtanyeongcheon-ro, Hwaseong-si, Gyeonggi-do, Republic of Korea
Japan distributor BIT DEVICE Co., Ltd., 6-3-24-L430 Nishinakajima, Yodogawa-ku, Osaka, Japan

Product page

Try it

INTELLIZ

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Display screens and values are the product page's illustrative reading example, not customer screens.
The reading log and reading-interval charts are concept diagrams drawn from the same example's values.
The factory digital twin is an example view of a virtual factory in the INTELLIZ Solution Experience, not customer data.
Accuracy depends on capture conditions and the display; the scope is set per target after on-site validation.